



SIMULATION OF SOUND WAVE TRANSMISSION IN HETEROGENEOUS SOLID MEDIA TO UNDERSTAND UPV OF CONCRETE

Ayush Tanwar(23110056) Avdut Vijay Thorat (23110339)

Under the supervision of Prof. Gaurav Srivastava | Department of Civil Engineering, Indian Institute of Technology, Gandhinagar

INTRODUCTION

Ultrasonic Pulse Velocity (UPV) testing is a widely used Non-Destructive Testing (NDT) method that evaluates concrete quality by measuring the travel time of an ultrasonic pulse through the material. Higher velocities indicate dense, high-quality concrete; lower velocities may suggest voids, cracks, or poor compaction.

Concrete's heterogeneous nature - comprising aggregates, mortar, interfacial transition zones (ITZ), and pores - makes interpretation complex. Waves experience scattering, reflection, and attenuation due to material property variations, causing the apparent velocity to deviate from the true material velocity. This project develops a mesoscale numerical model to simulate these effects.

RESEARCH OBJECTIVES

- Simulate ultrasonic wave propagation in a mesoscale concrete model explicitly representing aggregates and mortar phases.
- Compare simulated UPV with experimental measurements from lab-cast M30 concrete cubes to validate the model.
- Analyse how UPV deviates from theoretical wave velocity and assess its relationship with elastic parameters.
- Identify conditions where UPV provides accurate or misleading information on concrete quality.

EXPERIMENTAL RESULTS

150×150×150 mm M30 concrete cubes were prepared with ~41% coarse aggregate and 59% mortar by volume, cured for 28 days, then subjected to UPV testing.

Parameter	Value	Parameter	Value
Grade	M30	w/c Ratio	0.43
Cement (OPC 43G)	458 kg/m³	Fine Aggregate	602 kg/m³
Coarse Aggregate	1117 kg/m³	Max. Agg. Size	20 mm
Water	197 kg/m³	Cube Size	150 mm

UPV = Path Length / Transmission Time

43.9 μs

BLOCK 1 — TRANSMISSION TIME
UPV = 3416.85 m/s

39.8 μs

BLOCK 2 — TRANSMISSION TIME
UPV = 3768.84 m/s

Both cubes yield UPV values in the **good–excellent quality range** as per IS 13311, confirming well-compacted M30 concrete.

SIMULATION MODEL RESULTS (ANSYS)

A **150×150×150 mm mesoscale cube** was modelled in ANSYS with randomly dispersed spherical coarse aggregates (Ø 8–16 mm) inside a continuous mortar matrix. Transient explicit dynamics analysis simulated ultrasonic wave transmission; a velocity probe on the opposite face recorded the first-arrival time.

MORTAR MATRIX		COARSE AGGREGATE	
Young's Modulus	28,060 MPa	Young's Modulus	60,000 MPa
Poisson's Ratio	0.20	Poisson's Ratio	0.17
Density	2100 kg/m³	Density	2700 kg/m³
Vol. Fraction	74%	Vol. Fraction	26%

Effective density = 0.74×2100 + 0.26×2700 = 2256 kg/m³ |
Mesh: 2 mm element size

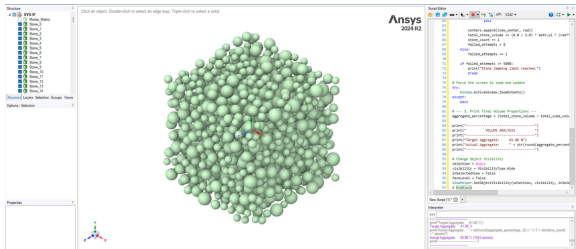


Fig 1: Coarse aggregate placement (Ø 8–16 mm)

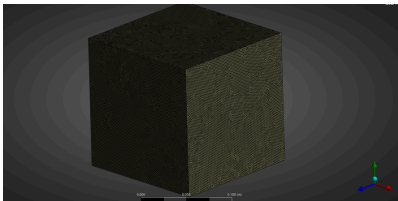


Fig 2: Meshed concrete cube

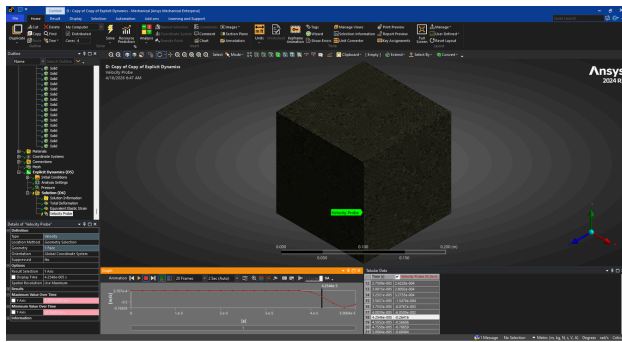


Fig 3: Simulated wave propagation result

First arrival time = 42.5 μs → $UPV_{sim} = 150 \text{ mm} / 42.5 \mu s = 3529.41 \text{ m/s}$

Simulated UPV falls within experimental range (3417–3769 m/s), validating the ANSYS mesoscale model.

YOUNG'S MODULUS ESTIMATION

Two approaches were used to estimate the effective Young's modulus of the concrete composite.

Hashin–Shtrikman (HS) Lower Bound — Since stiff aggregates are dispersed within a softer mortar matrix without forming a connected network, the HS lower bound is the most appropriate static homogenization model:

$$K = \frac{E}{3(1-2\nu)} \quad G = \frac{E}{2(1+\nu)}$$

$$K_a = 30,303.03 \text{ MPa} \quad G_a = 25,641.03 \text{ MPa}$$
$$K_m = 15,588.89 \text{ MPa} \quad G_m = 11,691.67 \text{ MPa}$$
$$V_f = 0.26 \quad V_m = 0.74$$

$$K^{HS-} = K_m + \frac{V_f}{\frac{1}{K_a - K_m} + \frac{3V_m}{3K_m + 4G_m}} \quad G^{HS-} = G_m + \frac{V_f}{\frac{1}{G_a - G_m} + \frac{6V_m(K_m + 2G_m)}{5G_m(3K_m + 4G_m)}}$$

$$E_{eff} = \frac{9K^{HS-}G^{HS-}}{(3K^{HS-} + G^{HS-})}$$

$$K^{HS-} = 18,424.2 \text{ MPa} \quad | \quad G^{HS-} = 14,208.4 \text{ MPa}$$
$$E_{eff} \text{ (HS-)} = 33.9 \text{ GPa} \quad | \quad \nu_{eff} = 0.193$$

Dynamic Modulus from UPV — Using the longitudinal wave velocity relation with $\rho = 2256 \text{ kg/m}^3$, $V = 3529.41 \text{ m/s}$, $\nu = 0.193$:

$$E = \rho \cdot V^2 \cdot (1+\nu)(1-2\nu) / (1-\nu)$$
$$E_{dynamic} = 25.52 \text{ GPa}$$

33.9 GPa

HASHIN-SHTRIKMAN
STATIC BOUND

25.52 GPa

UPV-BASED
DYNAMIC MODULUS

The ~25% gap arises because HS assumes perfect bonding and idealised geometry (static upper-bound stiffness), while the dynamic UPV modulus captures real effects — scattering, wave-path distortion, and microstructural discontinuities.

FUTURE WORK

- Increase aggregate volume fraction beyond the current 26% limit (typical real concrete ~40%) by improving geometry generation scripts in ANSYS SpaceClaim.
- Incorporate **Interfacial Transition Zones (ITZ)** as a distinct phase to capture their weakening effect on wave propagation.
- Extend simulations to include **voids and cracks** to study how defects alter UPV and assess NDT detectability limits.

CONCLUSION

- The mesoscale ANSYS model successfully reproduced the experimental UPV range (3416–3769 m/s) with a simulated value of **3529 m/s**, confirming model validity.
- Concrete heterogeneity - aggregate distribution, impedance contrast, and scattering - significantly influences UPV beyond just bulk elastic properties.
- The static Hashin–Shtrikman modulus (**33.9 GPa**) and UPV-based dynamic modulus (**25.52 GPa**) differ by ~25%, reflecting fundamentally different physical mechanisms.
- UPV should not be used as a direct indicator of material stiffness or compressive strength without accounting for wave physics and heterogeneity.
- Combining experimental UPV with numerical simulation provides a deeper, physically-grounded assessment of concrete quality.

REFERENCES

[1] Jiang et al., "Prediction of UPV for Cement, Mortar, and Concrete through a Multiscale Homogenization Approach," *Materials*, 2022. doi: 10.3390/ma15093241

[4] Chen et al., "FE simulation based on ultrasonic phased array," *IOP Conf. Series: MSE*, vol. 657, 2019.

[2] Darmon et al., "Modelling of ultrasonic velocity and attenuation for concrete inspection," *Ultrasonics*, vol. 154, 2025. doi: 10.1016/j.ultras.2025.107690

[5] Forti et al., "3D Mesoscale FE Modelling of Concrete under Uniaxial Loadings," *Materials*, vol. 13, 2020.

[3] Song et al., "Surface wave propagation in concrete using piezoelectric system," *Smart Mater. Struct.*, vol. 17, 2008.

[6] Sharifi et al., "Mechanical properties of concrete with aggregates and voids by mesoscale RVE," *Proc. ICE – Construction Materials*, 2025.